📊 Performance Test Results

FlightFinder - Test Execution Report

Test Date: [Date] | Tester: [Name]

1. Executive Summary

Metric	Result	Status
Overall Test Status	95% Pass Rate	✅ **PASS**
Average Response Time	1.2 seconds	✅ **PASS**
Peak Concurrent Users	120 users	✅ **PASS**
Error Rate	0.5%	✅ **PASS**
System Stability	Stable for 2 hours	✅ **PASS**

2. Detailed Test Results

2.1 Page Load Performance

Page	Target	Actual	Status
Landing Page	< 2s	1.5s	✅ PASS
Search Results	< 2s	1.8s	✅ PASS
Booking Page	< 2s	1.9s	✅ PASS
Dashboard	< 2s	1.4s	✅ PASS
Admin Panel	< 3s	2.3s	✅ PASS

2.2 API Performance

API Endpoint	Target	Actual	Status
POST /register	< 1s	650ms	✅ PASS
POST /login	< 1s	580ms	✅ PASS
GET /fetch-flights	< 1s	720ms	✅ PASS
POST /book-ticket	< 2s	1.3s	✅ PASS
GET /fetch-bookings	< 1s	850ms	✅ PASS

2.3 Database Performance

Query Type	Target	Actual	Status
User Lookup	< 200ms	120ms	✅ PASS
Flight Search	< 500ms	380ms	✅ PASS
Booking Insert	< 500ms	290ms	✅ PASS
Aggregate Queries	< 1s	750ms	✅ PASS

2.4 Load Test Results

Test Type	Users	Avg Response	Error Rate	Status
Baseline	5	450ms	0%	✅ PASS
Normal Load	50	1.1s	0.2%	✅ PASS
Peak Load	100	1.8s	0.5%	✅ PASS

Stress Test	150	2.9s	2.1%	⚠️ ACCEPTABLE
Endurance (2hr)	50	1.2s	0.3%	✅ PASS

3. Issues Identified

ID	Issue	Severity	Recommendation
P-001	Admin panel slow with 500+ records	Medium	Implement pagination
P-002	Response time degrades at 150+ users	Medium	Add caching layer
P-003	Memory usage increases over time	Low	Review for memory leaks

4. Recommendations

- **Implement Caching:** Use Redis for frequently accessed data
- **Database Indexing:** Add indexes on frequently queried fields
- **Code Optimization:** Optimize heavy database queries
- **CDN Integration:** Serve static assets via CDN
- **Load Balancing:** Prepare for horizontal scaling

5. Conclusion

Overall Assessment: The FlightFinder application demonstrates excellent performance under normal operating conditions. The system successfully handles 100 concurrent users with response times well within acceptable limits. Minor optimizations recommended for handling extreme load scenarios.

Recommendation: APPROVED FOR PRODUCTION with suggested optimizations to be implemented in next iteration.

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